MIA CARE

Create Your Telemedicine Platform

With Cloud-Native Technology

Abstract

Telemedicine is the new big thing as the advent of cloud-native technologies arrived. From Hospitals and Pharma to Insurancers, there is a need to create a digital architecture that embraces all levels of telemedicine. In particular, Italy is pushing healthcare providers to undertake digital transformation and equip themselves with microservices-based platforms by writing off a monolithic approach. The paper explores the benefits of adopting cloud-based technologies, along with the pillar services that conceive a modular architecture to bring to the market a digital platform that perfectly matches the telemedicine needs of individual healthcare providers.

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CHAPTER 01

What is Europe bringing to Digital Healthcare Innovation?

THE NEXT GENERATION EU

Although the Next Generation EU was created mainly as economic support in response to the Covid-19 crisis, it is a long-term strategic redesign tool for Europe according to a threefold directive: green transition, digitalization - mainly of public administration - and social inclusion.

The total investments, amounting to €806.9 billion divided into grants and subsidized loans, will be deployed in three main directions:

1. Single market, innovation, and digital - €11.5 billion
2. Cohesion, resilience, and values - €776.5 billion
3. Natural resources and environment - €18.9 billion

The second point is dedicated to the digitalization, modernization, and innovation of the health system, which aims at strengthening the resilience and cohesion between the EU Member States by investing in the green and digital transition, young people, health, and action to protect EU values. Here we find the allocation of resources dedicated to enhancing the transition to digital for the healthcare industry, thanks to the creation of the EU4Health program.

EU4HEALTH 2021 - 2027

EU4Health is a European Union (EU) Health Action Programme for the period 2021-2027 and, with a budget of €5.1 billion, represents the EU's response to the COVID-19 pandemic and will allow EU countries to access dedicated funding.
The program is designed to ensure a high level of quality in the delivery of health-related services in line with the 'One Health' approach and relies on four general pillars:

Disease prevention | Crisis preparedness | Health systems and health personnel | Digitalization of the health system

Among other topics discussed, the EU4Health Programme strongly aims to strengthen national health systems through better use of data, financing the development of new innovative digital tools and services, and stimulating the digital transformation of health to improve access to care.
Speaking specifically on telemedicine, COVID-19 accelerated the adoption of this innovative way to support patients.
However, some barriers limit the potential of telemedicine:

> **LEGAL UNCERTAINTY**
>
> There is a lack of shared guidance and a coherent policy framework within and across all Member States to overcome the lack of reimbursement and structured procedures to implement new digital assets in healthcare;

LIMITED ACCESS TO SERVICES

European citizens can access different levels of care quality due to the fragmentation that characterizes the EU Member States. This situation led to a poor diffusion and penetration of digital health services embracing both doctors and patients;

POOR DIFFUSION AND AWARENESS

There is still a lack of incentives for the deployment of telemedicine, digital health skills and literacy from all the stakeholders, therefore it is mandatory to conceive new channels of communication and create incentive programs to educate patients, administrative staff, and doctors.

THE ITALIAN LANDSCAPE

Mission 6 and National Telemedicine Platform

Tackling the COVID-19 pandemic was certainly one of the toughest challenges facing the Italian National Health Service. Despite the disastrous condition that brought the hospital system to the brink of exhaustion, the pandemic provided important indications of its future sustainability and efficiency by sparking a significant boost to new care delivery systems, such as telemedicine.

The European response to the pandemic crisis, given by the NextGenerationEU, envisages €68.9 billion in grants and €122.6 billion in loans for Italy. Resources to be managed within the perimeter of the National Recovery and Resilience Plan (NRPP) are divided into six Missions:

- Digitalization, Innovation, Competitiveness, Culture, and Tourism;
- Green Revolution and Ecological Transition;
- Infrastructure for sustainable mobility;
- Education and Research;
- Inclusion and Cohesion;
- Health.

Mission 6 of the Italian Recovery and Resilience Plan aims to renew the Italian health system. €15.63 billion is allocated to:

- Supporting the Italian healthcare system through the development of proximity networks, facilities, and telemedicine for territorial healthcare, with a budget of €7 billion;
- fostering innovation, research, and digitalization of the national health service, with a budget of €8.63 billion.

Focusing on the first expenditure item related to Local Assistance and Telemedicine, the goal is to strengthen the national health system by aiming at:

- Ensuring territorial proximity by overcoming the fragmentation among the different regional health systems and guaranteeing homogeneous access to care and the provision of Essential Levels of Care (LEA);
- improving prevention and territorial care by increasing the capacity to integrate hospital services, local health services, and social services to ensure continuity of care;
- promoting social equity in the vision of a "One-Health" approach;
- increasing the digital skills of the actors involved in the health system and promoting the evolution of hospitals into safe, technologically advanced, and sustainable facilities.

After closing an initial operational phase related to the urgent management of the pandemic peak, one of the first initiatives implemented by the Italian government was the creation of a tender for the construction of a national telemedicine platform.

The main objective of the national platform is to create a fundamental level of interoperability that guarantees common standards for the telemedicine services developed by the regions, enhancing what is already available in the local contexts and integrating or completing the portfolio of services.

The call for tenders envisages a significant change and follows the dynamics of digital transformation that have affected all the major strategic sectors of the country.

The development of the whole architecture will follow the Agile methodology and rely on a microservices-based architecture, guaranteeing high levels of computational performance and data interoperability, as well as flexibility and resilience abilities.

CHAPTER 02

What is Telemedicine?

Insight into the areas of intervention

It is appropriate to start with a clarification. Telemedicine is defined as the set of medical and digital processes that enable the treatment of a patient at a distance or, more generally, the provision of remote healthcare services. In particular, four macro processes characterize Telemedicine:

- Televisit and Teleassistance
- Teleconsultation
- Telemonitoring

TELEVISIT AND TELEASSISTANCE

Televisit refers to a medical act in which the doctors interact remotely in real-time with the patient and have the opportunity to exchange clinical data, medical reports, images, and audio-video, related to the patient in real-time. In addition, the medical history can be collected using the video call chat embedded in the video platform.

Patients can book an appointment through a central booking system (known as CUP) using the channel they prefer (e.g., email, telephone, SMS). Alternatively, the appointment is booked directly by the doctor specialist. The scheduled appointment becomes visible within the hospital's platform and the regional

health service record if integrated with the health structure. After booking the visit, the patient is provided, by email or SMS, with the access link to the portal where the personal profile resides to fill in the necessary forms related to the current data protection and informed consent regulations (Privacy and GDPR compliance).

At the same time, the doctor assesses the patient's clinical records from his medical history and may ask for questionnaires to be filled in and to attach any health documents that come from external sources. Finally, during the televisit, the doctor can perform real-time anamnesis while interviewing the patient and, at the end of the televisit, proceeds to send the report to the patient's health records. The patient can view these documents on his profile and FSE if integrated with the healthcare system of the structure.

Teleassistance, on the other hand, is when the remote medical service involves the caregiver more than the patient. The ultimate aim is to facilitate the programs of caregivers by helping them to carry out the scheduled and repeatable activities at home.

TELECONSULTATION

Teleconsultation refers to a task in which a doctor interacts remotely with one or more specialists to dialogue about a patient's clinical situation, based primarily on sharing clinical data, reports, radiological images, and laboratory tests.
The purpose of teleconsultation is to share medical choices concerning a patient by the specialists involved and it is also the modality to provide the specialist 'second opinion', which is important for the definition of the best therapy.

There are two types of teleconsultation:

SYNCHRONOUS

when the consultation request is accepted in real-time by the referring specialist, and the patient is also present at the teleconsultation

ASYNCHRONOUS

when the request is conveyed via the platform shared by the doctors and is not immediately processed but waits for a specialist to take it over.

TELEMONITORING

Telemonitoring helps gather, organize, and view remotely vital and clinical parameters continuously, employing sensors that interact with the patient (biomedical technologies with or without applied parts). Medical Devices are configured according to the doctor's instructions and constantly connected to the digital system. The information collected is dedicated to monitoring pathologies by making them available to healthcare specialists according to the therapy structure.

The purpose of Telemonitoring activities is to monitor the measured health parameters, allowing the detection of parameters with greater frequency and uniformity while reducing the need for the patient to perform outpatient checks in person.

Once doctors decide on the proper device to use, they configure the duration of monitoring and the rules on alarms based on shared guidelines.

The telemedicine system of the healthcare facility receives the data collected by the device and makes them available for analysis by the doctor. The doctor specialist conducts evaluations and decides whether a remote or in-person visit is necessary. Otherwise, regular telemonitoring keeps going.
At the end of telemonitoring, the doctor draws up a final clinical evaluation, and the outcome may indicate the resolution of the clinical situation or the need to update therapy.

CHAPTER 03

Why overcome a monolithic approach and leverage a microservice-based architecture

THE LIMITATIONS OF MONOLITHIC ARCHITECTURES AND THE NATIONAL GUIDELINES FOR CREATING TELEMEDICINE PLATFORMS

In monolithic architectures, processes are closely interconnected and run sequentially as a single service, meaning that a total downsizing of the application is required if a single process has bugs.

Adding, deleting, or improving a single function of the monolithic application is more complex because it could potentially affect the entire codebase. This complexity limits experimentation by restricting the implementation of new features.

Furthermore, the definitive confirmation that the use of monolithic architectures goes to an end is certified by the call for tenders for a national telemedicine platform promoted by the Italian Agency for Regional Health Services (AGENAS).

The development of this new digital architecture comprises four layers:

- The Core Services for integration with the central national systems envisaged for the digital transition process of the services provided by the PA;

- The Enabling Services constitute a set of services to which applications can be associated to facilitate adoption by local contexts;

- The Minimum Telemedicine Services, each consisting of a set of applications that integrate with each other and the existing systems in local contexts to enable the delivery of telemedicine services;

- The Data Interoperability Layer enables the collaboration of Regions, thanks to the microservices that guarantee orchestration in local contexts.

This type of architecture, multi-layered and ready to adapt to the specific needs of single Regions, is uniquely designed as a set of microservices that perform in-scope tasks and can be adopted as required or not.

THE BENEFITS OF CLOUD-NATIVE TECHNOLOGY AND THE USE OF MICROSERVICES TO BUILD NEW DIGITAL SOLUTIONS

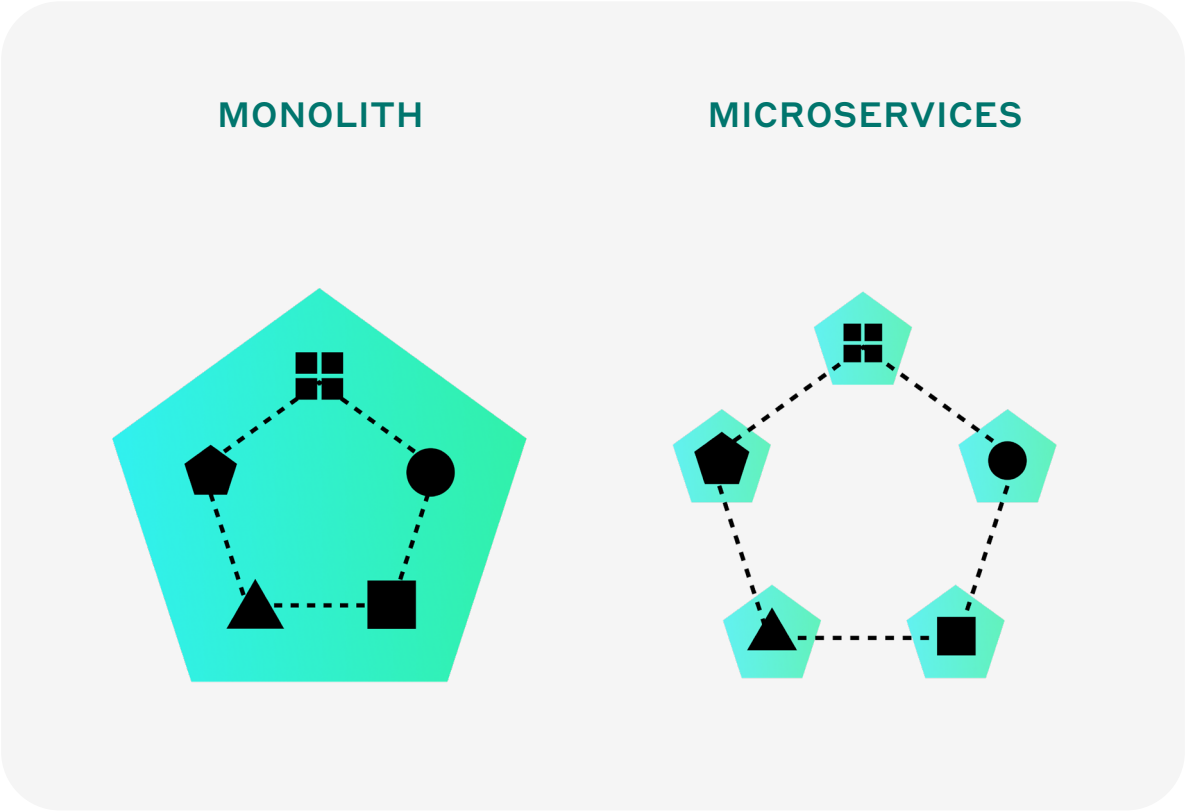
The use of cloud-native technologies makes it possible to address the new challenges that the ongoing digital transformation brings. While in several strategically important sectors, such as banking and transport, the value of digital is already clear to all stakeholders, and there are common legislative frameworks at the European level that ensure uniformity of intent, it is not the same for the healthcare market.

Nowadays, hospitals, pharmaceuticals, MedTech companies, and insurance companies operating in the healthcare world are taking steps to be ready for the incoming digitalization, but still, they need the right tools and skills to tackle digital challenges effectively. Leveraging a cloud-native approach, develop, deploy, and constantly update entire applications faster and more efficiently, hosting them in modern and dynamic environments such as public, private, and hybrid clouds.

The use of digital technologies relies on DevOps, based on Agile principles, and on the use of containers to streamline application development and enable the orchestration of workloads in a scalable manner. The role of modular microservices architectures, which enable faster and simplified updates, is fundamental but also the role of APIs, which allow different touchpoints to communicate flawlessly by guaranteeing the security of shared data.

Thanks to a microservice-based architecture, many independent components run together to make the service work, and all the communication between them and the legacy systems takes place via lightweight APIs. Since they run independently, each service can be monitored, updated, deployed, and scaled to meet the demand for specific functions of an application.

Since healthcare providers are focusing their strategy on delivering patient-centric services, it is easy to see how this solution is a winner, as it allows the patient journey to be tailored according to risk level, pathology, and chronicity.
In addition, it helps improve the territorial proximity thanks to the adoption of video streaming systems that can be easily integrated into a microservice-designed architecture.



CHAPTER 04

Mia-Care

The perfect solution for building an integrated and modular Telemedicine Platform

Mia-Care leverages the technological excellence of Mia-Platform to create digital services and products entirely dedicated to the world of healthcare. In particular, it equips healthcare providers with a modular platform that can be assembled according to their needs and business logic, and easily integrated with all central systems such as FSE, SPID, Tessera Sanitaria, and Digital Identity.

The solution focuses on the concept of composability, allowing the provider to choose autonomously which components to use for its telemedicine service.
The functional capabilities of the modules prepared by Mia-Care to set up a Telemedicine platform that includes all the process phases, from onboarding to Telemonitoring, are briefly outlined below.

TELECONSULTATION SERVICE

This service enables the creation and delivery of both tele-visit and teleconsultation services. The plug-in arose from the need to perform remote visits between doctor and patient, combined with the availability of other tools such as chat, whiteboard, and screen sharing. The Teleconsultation Service has been integrated within the Appointment Manager to allow the creation of remote visits and generate links to the teleconsultation that is visible in the calendar. Then for faster access, information is sent to doctors and patients directly via email or text messages.

USER MANAGER SERVICE

The User Manager allows the creation, deletion, and management of users accessing our portals and applications. It establishes the authorization of a user (doctor, patient, admin) to access dedicated resources and maintains the repository of all identities in the system. This module also allows the management of doctor and patient master data, as well as the creation of predefined groups (such as patient, doctor, admin, super admin) and the management of individual profiles for access to dedicated portals.

COMMUNICATION SERVICE

This plugin allows the integration of sending messages to different users via the following channels: email, SMS, and push notifications. Thus, visit reminders can be set up and sent, and individual patients or clusters of patients can be notified with specific messages concerning their treatment plans.

APPOINTMENT MANAGER

This module is designed for managing events in a calendar. It allows doctors to observe upcoming appointments and configure their availability, and patients to book a visit based on the doctors' availability.
Specifically, the Appointment Manager can be used for creating, updating, and canceling an appointment, as well as for automatically sending messages and reminders to the users involved in the event.

FHIR DATA SHARING

The FHIR (Fast Healthcare Interoperability Resources) data model is a standard that defines the format of medical information and its exchange via specific APIs. Thanks to the service developed by Mia-Care, doctors can compile reports and medical histories directly in FHIR format, and share any patient data using this protocol. The adoption of FHIR enables significantly faster sharing of health data with digital facilities that comply with the standard data model.

FHIR ADAPTER

Thanks to this plug-in, Mia-Care is already able to translate clinical data written in custom code into FHIR resources. This function is essential to ensure the correct interoperability of data, even from existing legacy systems, between the various local, regional and national telemedicine platforms.

THE REAL BENEFITS THAT THE ADOPTION OF MIA-CARE BRINGS

Seamless data integration and ease to scale

Fast configuration and automatic upgrades

Complete privacy and security compliance

Elevated UX and intuitive interfaces

Patient empowerment on therapy adherence

Full adaptability on your business needs

Conclusions

New cloud-native technologies and fifth-generation technology standards for broadband networks are the pillars on which innovative healthcare services can rely. During the Covid-19 pandemic, the importance of Telemedicine increased worldwide, and at the same time, the need to collect, organize and interoperate patient data to improve care quality. To project and build telemedicine solutions, healthcare organizations should analyze current processes and workflows, understand the pain points, and then focus on finding the right technology to leverage. Instead of merely buying stand-alone software, the wisest choice is to acquire a cloud-based software suite that provides customizable modules for single services that can be combined and integrated by maintaining high data integration and security standards. To achieve this goal, private and public providers rely on microservices-based applications that are perfect to face the dynamic needs of the healthcare industry and enable health structures to improve communication with internal systems of records and public databases.

MIA CARE

Mia-Care enables healthcare companies to rapidly build and scale a custom digital platform for developing secure, high-performing, and reliable digital health services anytime and everywhere